TITAN: Transformer-based Intelligent Task Allocation Network for Dual-Phase Learning and Optimal Task Scheduling

~~*Purpose, Approach, Findings, Impact.*~~

Abstract:
Traditional operating systems rely on fixed heuristics and manually crafted rules for task scheduling, limiting their adaptability to dynamic and diverse workloads. This paper presents a novel approach that leverages attention mechanisms and a two-phase learning strategy to address these limitations. We propose a hybrid model that initially learns scheduling behaviors through supervised learning by imitating existing schedulers, followed by reinforcement learning to optimize key performance metrics. Our architecture incorporates transformer encoders to capture contextual relationships among competing tasks, enabling more informed and adaptive prioritization decisions. Experimental results demonstrate that our attention-based scheduler reduces missed deadlines by 0.7% compared to the Earliest Deadline First (EDF) algorithm under complex workload conditions. Moreover, the reinforcement learning fine-tuning of TITAN effectively balances multiple scheduling objectives, including deadline adherence, waiting time reduction, and CPU utilization further reducing missed deadlines, average waiting and response time by 10.6%, 71.8% and 70% respectively. This work lays the foundation for intelligent, context-aware, adaptive, and learning-based resource management, offering a transformative alternative to traditional fixed-rule scheduling policies in a variety of domains requiring optimal task allocation, such as real-time systems, cloud computing, and distributed networks, offering a transformative alternative to traditional fixed-rule scheduling policies.